Project Name: MediVyapar

Tagline: Smart Solution For Ever Pharmacy

APP FLOW OVERVIEW (FROM OPENING THE APP)

1. Splash Screen

- MediVyapar Logo
- Tagline
- Loading animation

2. First Screen (Authentication)

When user opens app, they see:

- **Login**
- **New User Register**

REGISTRATION FLOW (FOR NEW OWNER)

When user clicks **Register**, app asks:

Basic Owner Details:

- Owner Name
- Mobile Number (OTP Verification)
- Email (optional)
- GST Number
- Pharmacy License Number

Auto Fetch Feature:

After GST & License submission:

- App automatically fetches:
    - Shop Name
    - Shop Address
    - Shop Location (GPS)
    - Store Type

(If auto-fetch fails → manual entry option)

AFTER REGISTRATION – INITIAL SETUP SCREEN

User sees 2 main options:

1. **Setup Shop**
2. **Setup Staff**

SETUP SHOP MODULE (CORE FEATURE)

Step 1: Scan Shop Structure

When owner clicks **Setup Shop**:

Wall & Rack Scanning

- App asks to capture pictures of:
    - All walls
    - All racks
    - All pigeon holes / boxes

AI Auto Structure Mapping

MediVyapar AI automatically:

- Detects racks
- Detects columns & rows
- Assigns smart location names like:

Example:

- Rack 1, Rack 2, Rack 3
- Rack1-A1, Rack1-A2
- Rack1-B1, Rack1-B2
- If small boxes: Rack1-A1-01, Rack1-A1-02

System generates a complete digital 3D/2D rack map.

Owner can:

- Edit location names if needed.

Step 2: Add Medicines to Locations

After structure setup:

App asks to scan medicines placed in each location.

Staff:

- Opens a location
- Takes picture of medicines in that pigeon hole

AI Recognition

Using AI:

- Detect medicine name
- MRP
- Batch Number
- Expiry Date
- Manufacturer
- Salt Composition

Store this with:

- Exact Rack/Box Location

After doing this with all location

Now system knows:
✓ What medicine
✓ Where it is
✓ Quantity
✓ Expiry
✓ Price

Shop becomes digitally mapped.

SETUP STAFF MODULE

Owner can:

- Add Staff Name
- Add Phone Number
- Assign Role & Permissions

Role Types:

- Staff (Limited Access)
- Manager (Medium Access)
- Owner (Full Access)

Staff Access Includes:

- New Sales

- Add Inventory
- Cannot see:
  - Sales Reports
  - Financial Data
  - Analytics
  - Multi-store settings

HOME DASHBOARD (AFTER COMPLETE SETUP)

Owner sees:

Dashboard Overview:

- Today's Sales
- Monthly Sales
- Profit Estimation
- Low Stock Count
- Expiry Alerts
- Fast Moving Medicines
- Slow Moving Medicines

Clickable sections open detailed reports.

MAIN OPERATIONS (AFTER SETUP)

Two Main Buttons:

1. **New Sale**
2. **Inventory Management**

NEW SALE MODULE (AI PRESCRIPTION READER)

When clicking **New Sale**:

Options:

- Scan Doctor Prescription
- Scan Old Medicine Strip
- Manual Search by Name

Prescription Scanning

- Camera opens
- Take photo of prescription
- AI reads doctor handwriting in milliseconds

App shows:

- List of detected medicines
- Quantity suggestion
- Alternative suggestions (if not available)

Shows Exact Location:

Example:

- Paracetamol 650mg → Rack2-B1-03

When owner picks medicine:

- Tap ✓ Tick (checklist style)

After all selected:

- Auto Generate Bill
- Option to apply Discount
- Print / WhatsApp Bill
- Auto reduce stock

If Medicine Not Available:

App automatically:

- Suggests alternative with same salt
- Suggests generic option
- Suggests nearby available stock (if multi-store enabled)

9️⃣ INVENTORY MANAGEMENT MODULE

When clicking **Inventory**, owner sees:

1. New Stock
2. Low Stock Alert
3. Expiry Alert
4. Stock Relocation Suggestion

9.1 New Stock

Two Options:

Option 1: Scan Supplier Invoice

- Take photo of invoice
- AI reads:
    - Medicine Names
    - Quantity
    - Batch
    - Expiry
    - Cost Price

App:

- Adds stock automatically
- Suggests exact location where to place medicines
- Checklist mode while placing

Option 2: Manual Add

- Scan medicine strip
  OR
- Select from dropdown

App:

- Updates quantity
- Suggests location

9.2 Low Stock Alert

Shows:

- Low stock medicines
- Out of stock medicines

Includes:

- Reorder Button

When clicking reorder:

- Generate WhatsApp formatted order
- Send to supplier

9.3 Expiry Alert

Shows:

- Medicines expiring in 30/60/90 days
- Expired medicines

Option:

- Mark returned
- Mark disposed
- Discount suggestion for near expiry

9.4 Stock Relocation Suggestion (AI Smart Feature)

AI analyzes:

- Fast moving medicines
- Slow moving medicines

Suggests:

- Move fast moving near counter
- Move slow moving to back racks

Goal:
Increase efficiency & reduce staff search time.

MULTI-STORE FEATURE

Owner can:

- Add multiple shops in profile

If multiple shops exist:

- On app open → choose store
- Separate dashboard for each store
- Separate stock
- Centralized analytics (combined view)

STAFF LOGIN FLOW

If staff logs in with phone number:

They get:

- New Sale
- Add Inventory

They CANNOT:

- See financial reports
- See total sales data
- Access multi-store settings
- Edit shop structure

ADDITIONAL IMPORTANT FEATURES (ADDED FOR COMPLETENESS)

These are important and should be included:

- **Offline + Cloud Hybrid System**
  - Works offline
  - Syncs automatically when internet available
- **Data Backup**
  - Daily auto cloud backup
  - Owner can export data in Excel
- **Theft & Leakage Detection**
  - Stock mismatch alerts
  - Suspicious activity detection
- **Search Optimization**
  - Search by:
    - Brand name
    - Salt name
    - Barcode
- **Barcode Support**
  - Scan barcode for fast billing
- **AI Analytics (Future Phase)**
  - Demand forecasting

- Auto reorder prediction
- Seasonal demand insights

BILLING FEATURES

- GST compliant bill
- Thermal printer support
- WhatsApp bill sharing
- UPI QR integration
- Cash / Card / UPI tracking

TECHNICAL REQUIREMENTS FOR DEVELOPER

- AI Handwriting Recognition Model
- OCR for invoices
- Image recognition for medicines
- Cloud database (AWS / Firebase)
- Secure authentication with OTP
- Role-based access control
- Scalable architecture
- Multi-store database separation

🎯 TARGET USERS

- Small & Medium Medical Stores (India)
- Stores struggling with:
  - Medicine searching
  - Expiry losses
  - Stock mismanagement
  - Staff dependency

🎯 CORE PROBLEM SOLVED

- Staff wastes time searching medicines
- Owner doesn't know exact stock

- Expiry loss
- Wrong medicine given
- No data analytics
- Manual billing errors

🚀 **FINAL GOAL OF MEDIVYAPAR**

Create India's First:

AI-Powered
Digitally Mapped
Smart Pharmacy Operating System

TECHNICAL ARCHITECTURE BLUEPRINT

SYSTEM OVERVIEW

MediVyapar will follow a:

Hybrid Architecture

- Mobile App (Android First)
- Cloud Backend
- AI Processing Layer
- Local Offline Storage
- Central Database

Architecture Type:

Client → API Gateway → Application Server → AI Services → Database → Analytics Engine

HIGH LEVEL ARCHITECTURE DIAGRAM (LOGICAL FLOW)

Mobile App (Flutter/React Native)
↓
API Gateway (HTTPS REST APIs)
↓
Application Backend (Node.js / Django)
↓

| AI Services Layer |

```
| - Prescription OCR Model          |
| - Invoice OCR Model               |
| - Medicine Image Recognition      |
| - Recommendation Engine           |
-------------------------------------
        ↓
Cloud Database (Primary)
        ↓
Analytics Engine
        ↓
Admin Dashboard
```

FRONTEND ARCHITECTURE (Mobile App)

Recommended Tech:

- Flutter (Best for Android-first India market)
  OR
- React Native

Modules:

1. Authentication Module

- OTP Login
- Role-based access

2. Shop Setup Module

- Camera Integration
- Rack Mapping UI
- Manual Edit Interface

3. Sales Module

- Prescription Scan
- Barcode Scan
- Search Engine
- Billing UI

4. Inventory Module

- Invoice Scanner
- Manual Entry
- Alerts Page

5. Dashboard Module

- Sales Graphs
- Low Stock
- Expiry
- Analytics

6. Multi-Store Selector

BACKEND ARCHITECTURE

Recommended Stack:

- Node.js (Express)
  OR
- Django (Python preferred for AI integration)

Core Backend Modules

1. Authentication Service

- OTP Verification
- JWT Token
- Role-based authorization

2. Shop Management Service

- Create Shop
- Rack Mapping Storage
- Multi-store linking

3. Inventory Service

- Add Stock
- Reduce Stock
- Batch Tracking
- Expiry Tracking
- Low Stock Detection

4. Sales Service

- Create Bill
- Update Inventory
- Apply Discount

- GST Calculation

5. Analytics Service

- Daily Sales
- Monthly Sales
- Fast/Slow Moving Analysis
- Relocation Suggestion Engine

AI SERVICES LAYER

This is the heart of MediVyapar.

Prescription OCR Engine

Purpose:

- Read handwritten doctor prescriptions.

Tech:

- TrOCR / CRNN Model
- Fine-tuned for Indian prescription dataset

Flow:
Image → Preprocessing → OCR Model → Medicine Extraction → Database Match

Invoice OCR Engine

Purpose:

- Read supplier invoice
- Extract:
    - Medicine name
    - Quantity
    - Batch
    - Expiry
    - Cost price

Tech:

- LayoutLM
- Tesseract (initial MVP)

Medicine Image Recognition

Purpose:

- Detect medicine from strip image

Tech:

- CNN Image Classification Model
- Label dataset with Indian brands

Recommendation Engine

A. Alternative Medicine Suggestion

If medicine unavailable:

- Match by salt composition
- Suggest generic version

B. Stock Relocation AI

Input:

- Sales frequency
- Rack distance from counter

Output:

- Suggest new placement

DATABASE DESIGN

Recommended:

- PostgreSQL (Primary)
- Redis (Caching)
- Firebase (Optional for Realtime Sync)

Core Tables

Users Table

- user_id
- name
- phone

- role
- shop_id

Shops Table

- shop_id
- owner_id
- GST
- license_no
- address

Rack Structure Table

- rack_id
- shop_id
- location_code (Rack1-A1-01)

Medicines Table

- medicine_id
- name
- salt
- manufacturer
- MRP

Inventory Table

- inventory_id
- medicine_id
- shop_id
- location_code
- quantity
- batch
- expiry_date

Sales Table

- sale_id
- shop_id
- total_amount
- payment_mode
- timestamp

Sale_Items Table

- sale_item_id
- sale_id
- medicine_id
- quantity
- price

OFFLINE + CLOUD ARCHITECTURE

Mobile App:

- Local SQLite database

When Internet Available:

- Sync with Cloud API

Conflict Resolution:

- Last Updated Wins
  OR
- Server Authoritative Model

MULTI-STORE ARCHITECTURE

Each Shop:

- Separate inventory records
- Separate rack mapping
- Separate analytics

Owner:

- Linked to multiple shop_ids

Data Isolation:

- Shop-based filtering in every API query

SECURITY ARCHITECTURE

- HTTPS (TLS)
- JWT Token Authentication
- Role-Based Access Control
- Encrypted Sensitive Data
- Audit Logs for:
    - Stock changes
    - Staff activity

BILLING ARCHITECTURE

- GST Calculation Engine
- Printable Thermal Format
- PDF Invoice Generator
- WhatsApp Integration API
- UPI QR Support

ANALYTICS ENGINE

Daily Cron Jobs:

- Low Stock Detection
- Expiry Detection
- Fast Moving Analysis
- Slow Moving Analysis

AI Forecast (Phase 2):

- Demand Prediction Model
- Seasonal Trend Detection

DEPLOYMENT ARCHITECTURE

Cloud Options:

- AWS
- Google Cloud
- Azure

Recommended:

Backend:

- AWS EC2 / Render

Database:

- AWS RDS (PostgreSQL)

Storage:

- AWS S3 (Images & invoices)

Monitoring:

- CloudWatch

SCALABILITY PLAN

Phase 1:

- Single region deployment

Phase 2:

- Load Balancer
- Microservices architecture

Phase 3:

- Kubernetes container deployment

PERFORMANCE OPTIMIZATION

- Redis caching for medicine search
- Indexed database columns
- Compressed image uploads
- Lazy loading analytics

FUTURE AI EXTENSIONS

- Voice Command Billing (Hindi Support)
- Auto Reorder Prediction

- Theft Detection via anomaly detection
- Computer Vision Rack Monitoring (CCTV integration)